

The Model That Lied

*A novel about machine learning,
and the art of not fooling yourself*

A story retelling of the lecture

*Introduction to Machine Learning: Fundamentals of Supervised Learning & Classifiers
Evaluation*

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Each chapter ends with a short box, What Léa learned, that sums up the ideas in plain terms. A cheat sheet of every formula and a map back to the lecture slides are at the end of the book.

Characters

Léa, a young data scientist at the Banque de l'Iroise, Brest

Madame Kerguelen, her mentor, sixty-two, who has been building models
since before “data science” had a name

Mathis and Inès, two interns

Jean-Yves Le Bihan, the bank's director

Yannick, Léa's school friend, a junior doctor at the hospital

The nephew, who memorized the practice sheet

Ninety-Nine Percent

It was raining in Brest, which was not news. Rain in Brest was less a weather event than a permanent resident, and Léa had long ago stopped owning an umbrella out of a kind of respectful surrender.

She burst into the office on the fourth floor of the Banque de l'Iroise with her laptop held open in front of her like a waiter carrying a cake.

"Ninety-nine percent," she announced.

Madame Kerguelen did not look up. She was sixty-two, had been building models since before the words "data science" existed, and drank her tea so strong it was practically a solid. "Ninety-nine percent of what?"

"Accuracy! The fraud detector. I trained it on last month's transactions. Ten thousand of them. It gets ninety-nine percent right."

Now Kerguelen looked up. Not impressed. Curious, which was worse.

"How many of those ten thousand were fraud?"

Léa checked. "A hundred."

"And how many of those hundred did your model catch?"

Léa opened her mouth, then scrolled. Then scrolled again. Her face did something complicated.

"...Zero."

Kerguelen nodded slowly, as if Léa had just confirmed the weather. "So your model looked at the world, noticed that ninety-nine out of a hundred transactions are honest, and decided the smartest strategy was to say 'honest' to everyone. And it was right ninety-nine percent of the time. It's not a fraud detector, Léa. It's a very expensive way of writing the word *no*."

. . .

Léa sat down heavily on the chair that nobody ever used because it wobbled.

“But accuracy is *the* metric. It’s the first thing everyone reports.”

“It is the first thing everyone reports,” Kerguelen agreed, “which is exactly why it fools so many people. Accuracy asks one question: what fraction of your answers were correct? It doesn’t ask *which* answers. And in the real world, the thing we care about is almost always rare. Fraud. Disease. A crack in an airplane wing. The interesting class is the small one.”

She pulled a notepad toward her and drew two little grids.

“Let me show you something worse. Imagine two models, tested on ten thousand transactions, a hundred and fifty of them fraudulent.”

She wrote beneath the first grid: *Model One*.

“Model One actually tries. It catches a hundred of the hundred and fifty frauds. It misses fifty. And to do that, it’s a bit jumpy: it wrongly flags a hundred and fifty honest customers, who will call us very annoyed. Everything else it gets right. Accuracy?”

Léa did the arithmetic in her head. “The hundred frauds it caught, plus the honest ones it left alone... nine thousand seven hundred. So nine thousand eight hundred correct out of ten thousand. Ninety-eight percent.”

“Good. Now Model Two.” Kerguelen wrote. “Model Two is lazy. It says ‘honest’ to everything. It misses all hundred and fifty frauds, and it never bothers a single honest customer.”

“So it’s right on the nine thousand eight hundred and fifty honest ones... ninety-eight and a half percent.” Léa stopped. “Wait. The lazy one wins?”

“By accuracy, yes. It cuts the error rate from two percent to one and a half. Any manager would tell you that’s a twenty-five percent improvement. And it’s completely useless, because it never, ever catches fraud.” Kerguelen tapped the pen on the second grid. “The *less* accurate model is the *more* useful one. Remember that. Accuracy isn’t wrong, Léa. It’s just not enough on its own. You’ll need other measures, ones that look specifically at how you do on the rare class. We’ll get to them.”

Model One	said fraud	said honest	Model Two	said fraud	said honest
real fraud	100	50	real fraud	0	150
real honest	150	9,700	real honest	0	9,850
accuracy		98.0%	accuracy		98.5%

Kerguelen's two grids. Green cells are correct answers, red cells are mistakes.

. . .

Léa stared at the grids for a while. Then she remembered something, and her stomach dropped a second time.

"There's another problem," she said quietly.

"There usually is."

"When I measured the ninety-nine percent... I measured it on the same transactions I trained on."

Kerguelen put her tea down. This, apparently, deserved both hands free.

"Let me tell you about my nephew," she said. "When he was fourteen, his maths teacher gave the class a practice sheet before the exam. My nephew memorized every answer on it. Word for word. Then he took the practice sheet as a test, and scored a hundred percent. He was very proud."

"And the real exam?"

"Different questions. He scored thirty-one." She shrugged. "He hadn't learned mathematics. He'd learned *that sheet*. And your model has done the same thing. When you measure it on its own training data, you're asking it questions it has already seen the answers to. Of course it looks good. Any number you compute on training data will be optimistic. Always."

"So I need different data to test it on."

"You need data it has *never seen*. A test set, locked away during training, only opened at the end. And even that's not the whole story, because next month's transactions won't look exactly like last month's. Fraudsters adapt. Seasons change. Christmas shopping looks nothing like February. The future is never quite the same as the past." She paused. "Which means you also need a *protocol*: a careful, honest procedure for how you split data, train, and test. Otherwise you're just finding creative

ways to lie to yourself.”

. . .

The rain had picked up. Léa watched it slide down the window in long, uncertain lines.

“So how do I know if a model is actually good?”

Kerguelen smiled for the first time that morning. “Now *that* is the right question. And it’s really three questions.” She counted on her fingers.

“First: how reliable are your results? How much can you trust what the model learned? Second: how should you measure performance? Which number actually reflects what you care about? And third: when you have several models competing, how do you compare them fairly?”

“That sounds like a lot.”

“It’s the whole job. Building models is the easy part now. Any student can train a model in four lines of code. Knowing whether to believe it — that’s where the craft is.”

She leaned back.

“And ‘good’ isn’t even one thing. A good pattern should be *valid*: accurate, robust, reliable on new data. It should be *useful*: it saves money, or time, or lives. Ideally it’s *novel*, it tells us something we didn’t already know. And it should be *understandable*, because if you walk into the director’s office with a model nobody can explain, he’ll throw you out, and he’ll be right to. The first two, you can measure with numbers. The last two are more a matter of judgment.” She waved a hand. “For now, we’ll focus on validity. It’s the foundation. Everything else is built on it.”

Léa closed her laptop. The ninety-nine percent was still in there somewhere, shining, meaningless.

“Where do I start?”

Kerguelen stood, walked to the whiteboard, and uncapped a marker that squeaked in protest.

“We start,” she said, “with a question you’ve never asked yourself. When your model makes a mistake, *where does that mistake come from?*”

She drew a large circle on the board, and outside it, a single dot.

“There are exactly three places an error can come from. One of them you can never fix. The other two are at war with each other. And every

decision you'll ever make as a data scientist is really about choosing a side in that war."

What Léa learned

Accuracy is the share of correct predictions, and on imbalanced data it can be high for a useless model. Error measured on training data is always optimistic. A trustworthy estimate needs test data the model never saw and an honest protocol. A good pattern is valid, useful, novel and understandable; validity comes first.

The Three Sources of Error

Kerguelen stood back from the whiteboard and let Léa look at it: a big circle, and outside it, a single dot.

“Before I explain this, I need you to understand what you’re actually doing when you train a model. Not the code. The *situation*.”

She wrote a large **P** in the corner of the board.

“Somewhere out there is the real process that creates our data. Every customer, every purchase, every fraudster on every rainy Tuesday. That process follows some pattern, some distribution. Call it P. Here’s the painful part: you will never see P. Nobody will. It’s too big, too messy, and it keeps changing.”

“Then what do I see?”

“A handful of examples it produced. Ten thousand transactions, each with a label: fraud or not fraud. That’s your sample, S. Everything you learn, you learn from S. And everything you want, you want on P, on the transactions that haven’t happened yet.” She tapped the P. “The error you care about is the probability that your model gets a *future* example wrong. That’s called the **generalization error**. The whole art is to make it small using only the tiny window S gives you.”

$$\text{generalization error of } h = \Pr_{(x, y) \sim P} [h(x) \neq y]$$

. . .

Léa frowned. “But if I only have S , how can I possibly know anything about transactions I haven’t seen?”

“Excellent. You can’t. Not without an assumption.” Kerguelen sat on the edge of the desk. “When my daughter was three, she saw a white swan at the lake in Quimper. Then another. Then eleven more. And she announced that all swans are white. Was she being stupid?”

“No. She was generalizing.”

“Exactly. And to generalize, she had to *assume* something: that the swans she hadn’t seen would resemble the ones she had. Every learner, human or machine, needs assumptions like that to leap from what it’s seen to what it hasn’t. We call them the **inductive bias**. Without them, learning is literally impossible. Ten thousand examples tell you nothing about the ten thousand and first unless you assume *something* connects them.”

She drew an arrow into the circle.

“And your biggest assumption is this circle. When you chose your model, you chose a family of possible functions it’s allowed to be. If you pick a linear model, the circle contains only straight lines. If you pick a decision tree, the circle contains only trees. This is the **hypothesis space**, H . Your model will be *something inside this circle*. It cannot be anything else.”

. . .

“Pedro Domingos once said every learning algorithm is really three things,” Kerguelen went on. “Think of it like cooking. First, the **representation**: which dishes are you even capable of making? That’s your circle. If a dish isn’t in your repertoire, no amount of effort will produce it. Second, the **evaluation**: the taste test, the rule that says this attempt is better than that one. Third, the **optimization**: how you search through all the dishes you *could* make to find the one that tastes best.”

“So linear regression is... lines, squared error, and solving an equation.”

“Now you’re cooking.”

“And there’s a promise we’d like every learner to make,” Kerguelen added. “Leslie Valiant wrote it down in 1984. He called it **probably approximately correct** learning — PAC, for short. Read the name

literally. *Approximately correct*: the model's error is below some small tolerance you choose. *Probably*: that happens with high probability, because you can always be unlucky with a strange sample. A PAC learner says: give me enough examples, and with high probability I'll give you a model with low error. And Valiant insisted 'enough' must be reasonable — a polynomial amount of data and computation, not an astronomical one." She smiled. "Years later, a man called Schapire asked whether learners that are only *slightly* better than a coin toss could be combined into a strong one. The answer was yes. It's called boosting. You'll meet it one day."

Léa pointed at the lonely dot outside the circle. "And that?"

"That's **f**. The truth. The best possible function, the one that would predict fraud as well as it can ever be predicted." Kerguelen paused. "And notice where I drew it."

"Outside the circle."

"Often, yes. The truth doesn't care what model you chose. It's under no obligation to be a straight line just because you like straight lines."

. . .

She drew a second dot, inside the circle, as close to **f** as it could get while staying inside.

"This is **h-star**. Imagine I could give you not one sample of ten thousand transactions but a hundred different samples, all from the same **P**. You train a model on each. A hundred slightly different models. Then you average their predictions. That average model is **h-star**. It's the best your circle can do *on average*, with the luck of the draw smoothed away."

Then she drew a little cloud of dots scattered around **h-star**.

"And these are the models you *actually* get. Each one trained on a particular sample. Yours, Léa, is just one of these dots. Call it **h**."

She stepped back. Three characters on the board: the truth, the ideal-within-your-circle, and your actual model.

"Now. Your model makes errors. Where do they come from?"

She drew a line from **f** to **h-star**.

"**One: bias**. The distance from the truth to the best thing your circle can offer. This isn't bad luck. It's baked in. You chose a circle that doesn't

contain the truth, so even with infinite data and perfect averaging, you'll miss. It's a *systematic* error, an approximation error. It's the price of your assumptions."

She drew a line from h^* to one of the scattered dots.

"Two: variance. The distance from the ideal average to the model you actually got. This one *is* luck. Your particular sample had its own quirks, and your model absorbed them. Train on a different sample and you'd get a different dot. Variance is an estimation error: it measures how much your model changes when the data changes."

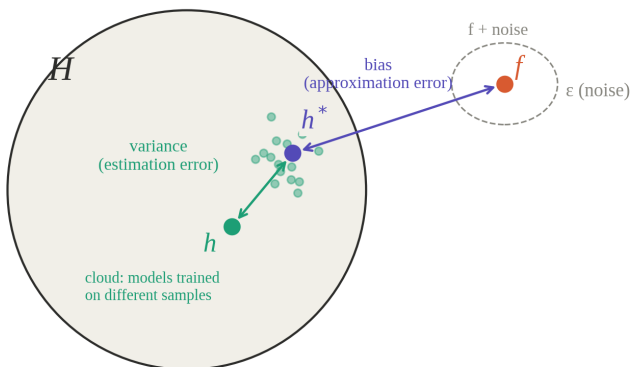
Finally she wrote a small Greek letter next to f : ϵ .

"Three: the incompressible error. Even the truth makes mistakes. Two customers can look absolutely identical in every column of our database, same age, same amount, same shop, same hour, and one is a fraudster and one isn't. No model can tell them apart, because the information simply isn't there. That's noise. You can't fix it. You can only accept it."

She underlined the whole thing.

$$\text{total error} = \text{bias} + \text{variance} + \epsilon$$

"You can't touch ϵ . So everything you do, every choice you make, is about the other two."



The circle is H , the family of models you allowed. The truth f may lie outside it. h^* is the best average model inside H , and each real sample gives a model h somewhere in the cloud around h^* .

. . .

Léa was scribbling notes. “Okay, but it’s still abstract. What does high bias actually *look* like?”

Kerguelen smiled like she’d been waiting for this. “Go and get Mathis and Inès.”

The two interns arrived looking worried, as interns do when summoned. Kerguelen opened a file: a few hundred points that rose, curved down, dipped below zero, and rose again. A gentle wave with noise sprinkled on top.

“This is our world. Mathis, you get twelve points chosen at random. Inès, twelve different random points. Same world, different samples. Each of you fits a straight line.”

They did. Mathis’s line sloped gently downward. So did Inès’s, almost identically.

“Look at them,” Kerguelen said. “Both lines cut straight through the wave, missing the peaks, missing the valley. They’re both *wrong*, in the same way. That’s high bias. A straight line simply can’t bend, so it can’t follow a wave, no matter which points you give it.” She turned to Léa. “But notice something else. Mathis and Inès got different data, and their lines are nearly twins. The model barely cares what sample it sees. That’s *low variance*. Stubborn, but consistent.”

“Now,” she said, “a degree three polynomial. Let your curves bend a little.”

The new curves rose, dipped, and rose, tracing the wave quite nicely. Mathis’s and Inès’s curves differed a little, but not much.

“Better. Now let’s get greedy. Degree ten.”

This time, the screens went mad. Each curve passed through nearly every one of its twelve points, snaking and twisting to touch them all. Mathis’s curve shot off the bottom of the screen near the right edge. Inès’s did something completely different. Between the points, both curves thrashed like a fish on a deck.

“My training error is almost zero,” Mathis said, sounding unsure whether to be proud.

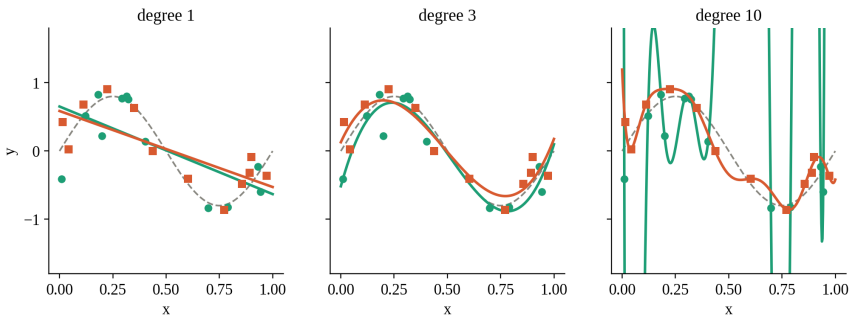
“Look at your coefficients,” Kerguelen said.

He scrolled. The numbers were in the hundreds of thousands, positive and negative, huge forces barely cancelling each other out.

“Your model has memorized your twelve points,” Kerguelen said. “Including the noise. Especially the noise. It fits them beautifully and understands nothing. Give it a new point from the wave and it’ll predict something absurd.” She pointed at the two wildly different curves. “And look: same world, same model, different samples, and you got completely different answers. That’s high variance. Low bias, because a degree-ten curve *can* bend into almost anything. But so flexible that it bends toward whatever random quirks your sample happened to have.”

Inès said quietly, “It’s your nephew. With the practice sheet.”

Kerguelen looked at her with new respect. “It’s exactly my nephew.”



Two samples from the same noisy wave (circles and squares), each fitted with polynomials of degree 1, 3 and 10. The dashed curve is the truth. Degree 1: both fits wrong in the same way (high bias, low variance). Degree 10: each fit chases its own points and the two disagree wildly (low bias, high variance).

...

That evening, Léa found herself at Le Tour du Monde, the little bar by the port, where there was an ancient dartboard and four regulars who took it far too seriously.

She watched them for a while and, somewhere around her second cider, understood.

Old Yann threw three darts and all three landed in a tight cluster, right in the bullseye. Low bias, low variance. The dream.

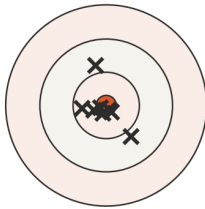
His brother Hervé threw three darts, tightly clustered, but high and to the left of the center, every single time. Consistent, but consistently wrong.

Something in his throw was simply off. High bias, low variance. The straight line.

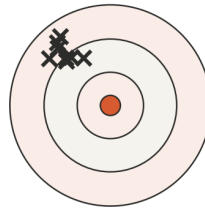
The young fisherman at the end threw darts all over the board. On average, they centered on the bullseye, but any single throw could land anywhere. Low bias, high variance. The degree-ten polynomial, a little drunk.

And the tourist who'd wandered in hit the wall twice and the board once, nowhere near the middle. High bias *and* high variance. Léa hoped she'd never build that model, but suspected she already had.

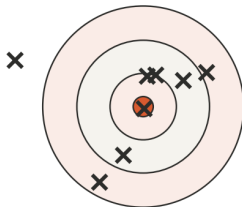
Yann
low bias, low variance



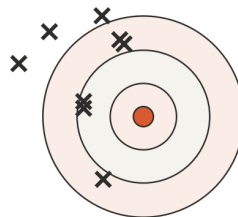
Hervé
high bias, low variance



the fisherman
low bias, high variance



the tourist
high bias, high variance



The four players at Le Tour du Monde. Bias is how far the centre of the throws sits from the bullseye; variance is how spread out the throws are.

. . .

The next morning, Kerguelen was waiting for her.

“So,” she said. “You’ve seen the war. Make your model too simple, and bias kills you: it can’t capture the truth. Make it too complex, and variance

kills you: it captures the noise instead. As you add complexity, your error on new data first falls, as bias shrinks, then rises again, as variance takes over. Somewhere in the middle is the bottom of the valley. Finding it is your job.”

“Too simple is underfitting. Too complex is overfitting.”

“Yes. And this is why learning is what mathematicians call an *ill-posed problem*. The data alone can never tell you the answer. Twelve points are consistent with a line, a cubic, and a degree-ten monster. You must choose. You must bring assumptions. And your error on the training set will lie to you about which choice was right, because the monster always wins on the training set.”

She took a small, worn book off her shelf. A history of medieval philosophy.

“Seven hundred years ago there was an English friar called William of Ockham. He had a rule: don’t multiply explanations beyond necessity. When two theories explain the facts equally well, prefer the simpler one.” She put the book back. “We call it Occam’s razor, and it’s served science well. For us it means this: if a simple model does about as well as a complex one, take the simple one. It’s faster to run. Easier to tune. Easier to explain to the director. And usually,” she tapped the board, right where the scattered dots were, “it has lower variance, so it generalizes better.”

“Is it always true? Simpler is always better?”

“No. Sometimes the truth really is complicated, and a simple model is just wrong. The razor isn’t a law of nature. It’s a good habit. A sensible default.” She capped her marker. “Start simple. Add complexity only when the evidence demands it.”

. . .

Léa sat down, and the chair wobbled, and for once she didn’t mind.

“There’s something I still don’t get,” she said. “You keep saying *error on new data*. But I don’t have new data. I’ve got S. The whole point is that P is invisible. So how do I actually *measure* the error I care about?”

Kerguelen’s eyes lit up.

“Now we’re getting somewhere. First we have to agree on what a mistake even *costs*. Is being off by one euro the same as being off by a

thousand? Then we'll talk about the difference between the error you *want* to know and the error you *can* compute." She turned the whiteboard over to its clean side. "They are not the same thing, and the gap between them is where careers go to die."

What Léa learned

We want low error on the unseen distribution P but only see a sample S , so every learner needs an inductive bias, above all the choice of hypothesis space H . Learning is representation + evaluation + optimization. Error splits into bias (H cannot reach the truth f), variance (the model depends on the particular sample) and irreducible noise ϵ . Simple models underfit, complex ones overfit; prefer the simplest model that does the job (Occam's razor).

The Price of a Mistake

The clean side of the whiteboard had a coffee stain on it that had survived eleven years of cleaning products. Kerguelen wrote around it, as she always did.

“Forget fraud for a moment,” she said. “The director has asked you for something else as well. Next month’s total card spending in each branch. A number, not a yes or no. That’s regression. Suppose the truth is ten thousand euros and you predict nine thousand nine hundred. How bad is that?”

“Not bad. A hundred euros off.”

“And if you predict five thousand?”

“Terrible.”

“So a mistake has a *size*, and we need a rule that turns each mistake into a number. We call it a **loss function**.” She wrote two conditions. “It only has to obey two laws. If you predict exactly the right answer, the loss is zero. If you predict anything else, the loss is positive. That’s all. Everything else is a choice, and the choice says something about what *you* care about.”

$$\begin{aligned}\ell(y, y) &= 0 \\ \ell(y, y') &> 0 \text{ whenever } y \neq y'\end{aligned}$$

. . .

“The simplest choice,” she went on, “is the **absolute loss**. Off by a hundred euros? Your loss is a hundred. Off by five thousand? Five

thousand. Fair and linear.”

“And the other kind?”

“The **squared loss**. You square the difference. Off by ten, loss one hundred. Off by a thousand, loss one million.” She let that sit. “Squared loss *hates* big mistakes. It would rather be slightly wrong everywhere than badly wrong once. That’s wonderful when big mistakes are catastrophic. It’s dangerous when your data has a few crazy values — one branch with a data-entry error of a billion euros — because the model will twist itself into knots to please that one point.”

$$\begin{aligned}\text{absolute loss: } \ell(y, y') &= |y - y'| \\ \text{squared loss: } \ell(y, y') &= (y - y')^2\end{aligned}$$

Léa thought about it. “And for fraud? There’s no ‘distance’ between fraud and not fraud.”

“For classification, the simplest loss is almost insultingly simple. Wrong: one. Right: zero. The **zero-one loss**.” Kerguelen drew a little staircase. “Now here’s something lovely. Take the zero-one loss and average it over your data. What do you get?”

Léa worked it out. “The fraction of mistakes. The error rate.” She blinked. “One minus accuracy. We’re back where we started.”

“We are. Accuracy is just the average zero-one loss wearing a nicer suit. And now you can see its weakness in a new way: the zero-one loss charges *the same price* for missing a fraud as for bothering an honest customer. Neither the bank nor the customer agrees.”

“There are other losses for classifiers that give scores instead of answers,” she added. “Suppose we code fraud as plus one and honest as minus one, and the model gives a score between minus one and plus one. The **linear cost** is one half of one minus the label times the score. Confidently right, you pay nothing. Confidently wrong, you pay one. Unsure, a score near zero, you pay a half. It rewards the model for being right *and* for knowing how sure it is.” She shrugged. “There are dozens of others. Researchers invent new ones every year. Pick the one that matches what a mistake actually costs you.”

zero-one loss: $\ell(y, y') = 1$ if $y \neq y'$, else 0
 linear cost: $\ell(h(x), y) = \frac{1}{2} (1 - y \cdot h(x))$, $y \in \{-1, +1\}$

. . .

“So,” Kerguelen said, “we have a price for each mistake. What we really want is the *average* price, over every transaction the bank will ever see. Every one that P will ever produce. That average is called the **risk** of your model. Or, if you like, the **true risk**. It’s just a more precise name for the generalization error.”

$$R_p(h) = E_{(x, y) \sim P} [\ell(h(x), y)]$$

“And h-star from yesterday...”

“...is the model in your circle with the lowest true risk. Yes. The whole game is finding it.” Kerguelen tapped the P. “But you already see the problem.”

“P is invisible. So I can’t compute the true risk.”

“Never. Not once, not for any model, not in your whole career. So we do the only thing we can. We compute the average loss on the data we *have*. That’s the **empirical risk**.”

$$R_S(h) = (1/n) \sum_{i=1..n} \ell(h(x_i), y_i)$$

“The true risk is what you want. The empirical risk is what you can compute. And now I’ll tell you why the gap between them is so treacherous.”

. . .

She pointed out the window, toward the port.

“Suppose I want the average height of every adult in France. I can’t measure sixty-eight million people. So I walk into a bar, measure the twenty people inside, and take their average. Is that reasonable?”

“If the bar is random, sort of. It’s an estimate.”

“Right. For a *fixed* bar, chosen without looking, the average of twenty people is an honest estimate of the country. It might be a bit high or a bit

low, but it isn't *systematically* wrong. Now suppose I visit a hundred bars, and I report the one with the tallest customers. Say, the one where the Brest basketball team drinks after training."

Léa laughed. "Then your average is nonsense. You picked the bar *because* it was tall."

"Exactly. And that is what training does." Kerguelen turned back to the board. "The empirical risk of a model you picked at random would be an honest estimate. But you didn't pick your model at random. You searched through the whole circle and chose the one with the *lowest* empirical risk on S. You went looking for the basketball team. So of course the empirical risk of your chosen model is too low. It's the tallest bar in town. It tells you how well the model fits S, not how well it will do on P."

"That's the nephew again."

"The nephew is everywhere, Léa. Once you've seen him, you'll never stop seeing him."

. . .

Mathis knocked and put his head round the door. "Sorry — Inès and I both trained fraud models last night. Hers says fraud on some transactions where mine says honest. Does that matter?"

Kerguelen's face lit up in a way that made Mathis visibly nervous.

"It matters enormously. Tell me: on this morning's transactions, the new ones nobody has labeled yet, how often do your two models disagree?"

Mathis checked. "About eight percent of the time."

"Then you've just learned something without a single label. Whenever you disagree, *at least one of you is wrong*. So at least one of your models makes mistakes on a good few percent of transactions, whatever your training scores say." She turned to Léa. "You can compare two models even on unlabeled data, by measuring how often they disagree. The true disagreement is the expected loss between their two predictions over P. You estimate it on a fresh set of inputs. Labels not required."

$$R_p(h, h') = E_{x \sim P} [\ell(h(x), h'(x))] \\ \text{estimated by } (1/n) \sum_i \ell(h(x_i), h'(x_i)) \text{ on new inputs}$$

“But notice what made it work,” she said. “New inputs. Data neither model had seen. Every honest estimate we’ve found this morning needs data the model didn’t train on.”

. . .

Kerguelen opened the bottom drawer of her desk. Inside, under a stack of old conference badges, was a small grey USB stick with a strip of masking tape on it. On the tape, in her neat handwriting, one word: **TEST**.

“Twenty percent of last year’s transactions,” she said. “Chosen at random, eleven months ago. No model in this building has ever been trained on them. No one has even looked at them, except to check the file opens.” She put it on the desk between them like a chess piece. “This is the only way to estimate the true risk honestly. Data the model has never seen, used once, at the end.”

“Why only once?”

Kerguelen smiled. “That,” she said, “is tomorrow’s lesson. And it’s about all the clever ways people break into the vault without realizing they’ve done it.”

What Léa learned

A loss function ℓ prices each mistake: zero when right, positive when wrong. Absolute and squared losses for regression (squared punishes big errors); zero-one loss for classification, whose average is the error rate. The true risk is the expected loss over P and can never be computed; the empirical risk is the average loss on the sample, and it is optimistic for the model you chose because you chose it for having a low value. Two models can be compared by how often they disagree, even without labels. Honest estimates always need unseen data.

The Vault

The next morning Léa arrived early, determined to do everything properly. She shuffled her ten thousand transactions, kept two thirds for training and put the last third aside as her own little test set. She trained, then tested. Error rate on the test third: 2.4 percent.

She was writing the number on a sticky note when Mathis leaned over her shoulder.

“I did the same split last night. Two thirds, one third. I got 1.9.”

“Same data? Same model?”

“Same everything. Different random shuffle.”

Kerguelen, who had apparently been listening from behind her newspaper, lowered it.

“Welcome to the **holdout**,” she said. “You hold out part of the data, you train on the rest, you test on what you held out. It’s honest, and it’s the simplest protocol there is. But you’ve just discovered its weakness. Your test set is only a sample too. A different third gives a different number. So your *estimate* of the error has its own variance.”

“So which one is right, 2.4 or 1.9?”

“Neither. Both. Repeat the split ten times with ten different shuffles and look at all ten numbers. Their average is a better estimate than any one of them, and their spread tells you how much to trust it. Never report a single split as if it were carved in granite.”

. . .

By the afternoon, Léa had moved on to a decision tree, and she wanted to know how deep to let it grow. She tried depth three, then five, then eight, then twelve, then twenty. Each time, she checked the error on her test third. Depth eight gave the lowest number. She wrote it down proudly: 1.7 percent.

Kerguelen read the note, then read Léa’s screen, then put down her tea.

“How many times did you look at the test set?”

“Five. One for each depth.”

“And you picked the depth with the best test score.”

Léa felt the familiar sinking. “The tallest bar in town.”

“The tallest bar in town. Every time you look at the test set and make a *decision* because of what you saw, a little of the test set leaks into your training. Look at it enough times and it isn’t a test set any more. It’s just more training data with a flattering name.” She held up the grey USB stick. “That’s why this one only gets opened once.”

“Then how am I supposed to choose the depth?”

“With a third pile.” Kerguelen drew three boxes. “**Train** on the first. **Tune** on the second — try all your depths, pick the best one there. Some people call it the validation set. And only when every decision is finished, you measure once on the third, the **test**. Train, tune, test. The tuning set is where you’re allowed to be greedy. The test set is where you’re honest.”



Three ways to split data. Holdout keeps one part aside for testing. Train, tune, test adds a separate pile for choosing settings. k-fold cross-validation (five folds shown) rotates the test part so that every example is tested exactly once.

. . .

“But now my training pile is even smaller,” Léa said. “I’m throwing data away twice.”

“Then stop throwing it away.” Kerguelen erased the three boxes and drew a long bar cut into ten slices. “**Cross-validation.** Cut your data into ten folds. Train on nine, test on the tenth. Then rotate: train on a different nine, test on the one left out. Do it ten times, so every fold is the test set exactly once. Every transaction gets tested exactly once, and every transaction is used for training nine times. You get ten scores. Report the average and the spread.”

“Why ten?”

“Tradition, mostly, and experience. Ten folds gives a good balance between the size of each training set and the cost of training ten models. Five is common too. And if you’re rich in computers, you can repeat the whole ten-fold procedure with different shuffles. That’s repeated cross-validation, and it’s even steadier.”

Inès, who had been silently eating a crêpe at the next desk, asked, “What’s the most folds you can have?”

“As many as you have examples. With n examples, you do n rounds: train on everything except one example, test on that single one. It’s called **leave-one-out.**” Kerguelen wrinkled her nose. “On ten thousand transactions, that’s ten thousand models. My cousin at the hospital uses it, though. She has forty patients in her study. When every example is precious, you don’t want to hold out a single one more than you have to.”

. . .

Léa set up ten-fold cross-validation and ran it. The first fold’s error rate came back: zero percent. The second: zero. The third: zero.

“Something’s wrong,” she said. “It’s perfect.”

Kerguelen came and looked. “Open the file. Show me the first rows.”

The file was sorted by label. Nine thousand eight hundred and fifty honest transactions, and then, at the very bottom, all hundred and fifty frauds.

“Your first folds contain no fraud at all,” Kerguelen said. “Your model says ‘honest’ to everything, and on those folds, it’s right every time. The

nephew strikes again.” She tapped the screen. “This is why the details of splitting matter as much as the idea. Four things. The *size* of your train and test parts. *Shuffle* before you cut, because files are almost never in random order. *Stratify*: with only a hundred and fifty frauds across ten folds, a random cut could give one fold five frauds and another twenty-five, so you make sure every fold keeps the same proportion of frauds as the whole dataset. And *control the shuffle*: fix the random seed, so that tomorrow you — or someone checking your work — can get exactly the same folds.”

“Mathis, you’ll like that last one,” Léa said. “You couldn’t reproduce your 1.9 this morning.”

Mathis went pink and fixed his seed.

“One more thing,” Kerguelen added, more quietly, “and it isn’t on anyone’s slides. Fraud changes over time. If you shuffle January and December together, your model trains on December to predict January. It gets a little peek at the future, and the future is exactly what it won’t have in real life. For data like ours, I also test the honest way: train on the past, test on the months that came after. Shuffling is the default. It’s not a religion.”

. . .

That evening Kerguelen drew a graph Léa would see in her sleep for years.

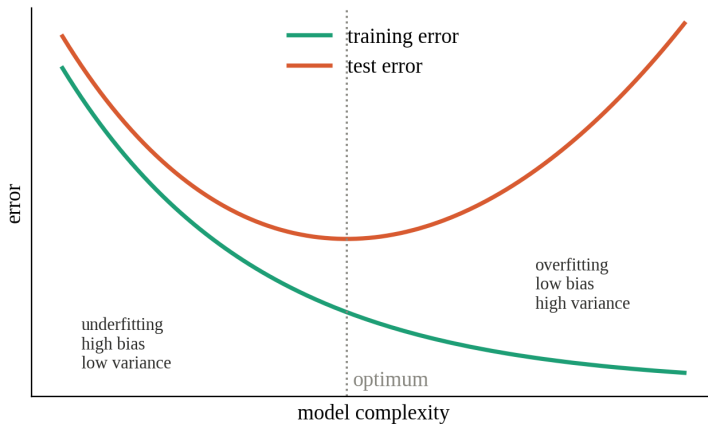
Along the bottom, model complexity: tree depth, polynomial degree, whatever dial makes the model more flexible. Up the side, error. Two curves.

“This one is the error on the training data,” she said, tracing a line that fell and fell and fell, down towards zero. “Give the model more freedom and it always fits the training data better. Always. This line never turns around.”

The second curve fell too, at first, alongside the first. Then it slowed, flattened, and turned back upward, climbing away while the training curve kept sinking.

“This is the error on data it hasn’t seen. On the left, both errors are high: the model is too simple, high bias, **underfitting**. On the right, the training error is tiny and the test error is climbing: the model is memorizing, high variance, **overfitting**. And there,” she tapped the bottom

of the second curve, “is the complexity you’re looking for. The gap between the two curves is how much your model is lying to you.”



As complexity grows, training error keeps falling while test error falls and then rises. Left of the optimum the model underfits (high bias); right of it, it overfits (high variance).

“So overfitting has a precise meaning?”

“It does. Your model h overfits if there is another model h' in the same circle that does *worse* on the training data but *better* on the real distribution. Better on the practice sheet, worse on the exam. That’s the definition. And you can only detect it with data the model hasn’t seen.”

h overfits if there exists $h' \in H$ with $\text{error}_{\text{train}}(h) < \text{error}_{\text{train}}(h')$ and $\text{error}_p(h) > \text{error}_p(h')$

...

“One last question,” Léa said, packing up. “Which model should I use? Trees? Logistic regression? Those support vector machines everyone talks about? Surely one of them is just the best.”

Kerguelen laughed, properly, for the first time that week.

“Two mathematicians, Wolpert and Macready, proved something wonderful and very annoying. Averaged over *every possible problem*, every learning algorithm performs exactly the same. For every problem where trees beat logistic regression, there’s another where logistic regression wins by just as much. They called it the **No Free Lunch**

theorem.”

“So there’s no best model.”

“There’s no best model *in general*. There’s only the best model for *your* problem, with *your* data, judged by *your* measure. Which is why you must measure honestly.” She switched off the light. “And speaking of your measure. Tomorrow, we go back to where we started. The confusion that made you so happy on Monday.”

What Léa learned

Holdout: train on part (often two thirds), test on the rest, and repeat because one split is noisy. Choosing settings on the test set leaks it, so use train, tune and test piles. k-fold cross-validation (usually $k = 10$) tests every example once and gives an average and a spread; leave-one-out is the extreme case for tiny datasets. Shuffle, stratify, fix the seed, and respect time order when it matters. Training error always falls with complexity; test error has a U shape. There is no universally best algorithm (No Free Lunch).

Four Boxes

On Thursday morning there was a new grid on the whiteboard, drawn with a ruler, which meant Kerguelen considered it important.

“Monday’s two models,” she said. “Model One, the one that tries. I want every number it produced, in one place.”

Léa filled it in. Rows for what each transaction really was, columns for what the model predicted.

	<i>predicted fraud</i>	<i>predicted honest</i>	<i>total</i>
<i>really fraud</i>	TP = 100	FN = 50	150
<i>really honest</i>	FP = 150	TN = 9,700	9,850
<i>total</i>	250	9,750	n = 10,000

Model One’s confusion matrix. Rows are the truth, columns are the model’s answer.

“This is a **confusion matrix**,” Kerguelen said. “It’s the most important table in this building. Almost every measure you will ever use is made of these four boxes. Name them.”

“Top left: frauds we called fraud. **True positives**. A hundred.”

“Why ‘positive’?”

“Because fraud is the class we’re hunting. ‘Positive’ is what the model *said*. ‘True’ means it was right.”

“Good. Carry on.”

“Top right: frauds we called honest. **False negatives**. The model said negative, and it was wrong. Fifty. Bottom left: honest customers we called fraud. **False positives**. A hundred and fifty. Bottom right: honest

customers we left alone. **True negatives.** Nine thousand seven hundred.”

“Now accuracy.”

“The diagonal over the total. The true positives plus the true negatives, divided by n. Ninety-eight percent. And the error rate is everything off the diagonal: two percent.”

“And now you see why accuracy fooled you. Nine thousand seven hundred of those ninety-eight hundred correct answers come from the honest customers. The box we care about, the frauds, is drowned out by the box we don’t.”

$$\begin{aligned}\text{accuracy} &= (TP + TN) / n = 1 - \text{error rate} \\ \text{error rate} &= (FN + FP) / n\end{aligned}$$

. . .

At ten o’clock, the director, Monsieur Le Bihan, came down to ask how the fraud project was going. Kerguelen asked him a question instead.

“Jean-Yves, what does a missed fraud cost us?”

“On average? About two thousand euros. Chargebacks, investigation, the lot.”

“And a false alarm? An honest customer whose card we block?”

He thought about it. “A phone call from the fraud team. Twenty euros, say. Plus an irritated customer.”

When he had gone, Kerguelen drew a second grid next to the first. “A **cost matrix**. Same shape. Instead of counts, the price of each kind of outcome. Getting it right costs nothing. A missed fraud: two thousand. A false alarm: twenty. Errors do not all cost the same, and the matrix isn’t symmetric. Now, Léa, what did each model cost us?”

Léa multiplied. “Model One missed fifty frauds, so a hundred thousand euros. Plus a hundred and fifty false alarms, three thousand. A hundred and three thousand in total.”

“And the lazy one? Ninety-eight and a half percent accurate?”

“It missed all hundred and fifty frauds.” She stared at the result. “Three hundred thousand euros.”

“The model with the worse accuracy saves the bank nearly two hundred thousand euros a month,” Kerguelen said. “If you have a cost matrix, use

it. The trouble is, you rarely do. Jean-Yves said twenty euros for a false alarm. But what if the irritated customer closes his account and takes his mortgage elsewhere? Nobody really knows what that costs. So we also need measures that don't depend on prices nobody can agree on."

. . .

"Two questions," Kerguelen said. "Ask them about Model One, and you'll never need to memorize the formulas.

"First question: *when the model raises an alarm, how often is it right?* It raised two hundred and fifty alarms — a hundred real frauds and a hundred and fifty false ones. So when it cries fraud, it's right a hundred times out of two hundred and fifty."

"Forty percent."

"That's **precision**. The true positives divided by everything the model *called* positive. It measures *quality*. The fraud team investigates two hundred and fifty alerts, and a hundred are real. High precision means few false positives: the team isn't wasting its days chasing honest people."

"Second question: *of all the frauds that really happened, how many did the model catch?* There were a hundred and fifty. It caught a hundred."

"Sixty-seven percent."

"That's **recall**. The true positives divided by everything that *really* was positive. It measures *quantity*: how much of what you were looking for you actually found. High recall means few false negatives: few frauds slip past."

$$\text{precision } p = TP / (TP + FP) = 100 / 250 = 0.40$$

$$\text{recall } r = TP / (TP + FN) = 100 / 150 \approx 0.67$$

Léa frowned. "Why can't I just maximize both?"

"Try it. How do you get a recall of a hundred percent?"

"Flag... everything." She saw it immediately. "Every fraud is caught. But I've flagged ten thousand transactions to find a hundred and fifty. Precision one and a half percent. The fraud team would quit."

"And a precision of a hundred percent?"

"Only flag the one transaction I'm absolutely certain about. If I'm right, precision is perfect. And recall is one in a hundred and fifty."

“So they pull against each other. Search engines have lived with this for decades. They call the things you want ‘relevant documents’. Precision asks: of the documents I returned, how many were relevant? Recall asks: of all the relevant documents, how many did I return? Returning the entire internet gives perfect recall. Nobody would call it a good search engine.”

. . .

“Then I need one number that balances them,” Léa said.

“The **F1-measure**. The harmonic mean of precision and recall.” Kerguelen wrote it. “For Model One, two times point four times point six seven, over point four plus point six seven. About point five.”

$$F_1 = 2 \cdot p \cdot r / (p + r) = 2 \cdot TP / (2 \cdot TP + FN + FP) \approx 0.50$$

“Why harmonic? Why not just the average?”

“Because the ordinary average forgives too much. Take the model that flags one transaction and gets it right. Precision one, recall almost zero. The ordinary average is about a half, which sounds respectable. The harmonic mean is nearly zero, which is the truth. F1 is only high when *both* precision and recall are reasonably high. It won’t let one of them hide behind the other.”

“Look at the second formula, too,” she added. “True positives, false negatives, false positives. What’s missing?”

“The true negatives.”

“F1 ignores the nine thousand seven hundred honest customers we correctly left alone. On purpose. That’s exactly the box that inflated your accuracy.”

“But for fraud, I care more about catching frauds than about false alarms.”

“Then weight it. The general form is the **F-beta measure**. Beta tells it how many times more you care about recall than precision. F2 weighs recall more — good for fraud, or for a cancer screening, where a miss is the disaster. F0.5 weighs precision more — good for a spam filter, where throwing away a real email from your mother is worse than letting one advertisement through.”

$$F_{\beta} = (1 + \beta^2) \cdot p \cdot r / (\beta^2 \cdot p + r)$$

$$F_2 = 5pr / (4p + r)$$

$$F_{0.5} = 1.25pr / (0.25p + r)$$

...

That evening Léa had dinner with her old school friend Yannick, now a junior doctor at the hospital. Over galettes she described her week.

“Precision, recall...” Yannick shook his head. “We use the same boxes. Different names. Your recall is what we call **sensitivity**: of all the sick patients, what fraction does the test detect? If sensitivity is ninety percent, ninety percent of the sick group are flagged as sick.”

“And the honest customers? The healthy ones?”

“**Specificity**: of all the healthy people, what fraction does the test correctly call healthy? True negatives divided by all the real negatives. If specificity is eighty percent, eighty percent of healthy people get a clean result, and twenty percent get a false alarm. That twenty percent is the **false positive rate**. The false alarm rate. It’s just one minus specificity.”

Léa calculated on a napkin. “My model: nine thousand seven hundred honest customers left alone, out of nine thousand eight hundred and fifty. Specificity ninety-eight and a half percent. False positive rate, one and a half.”

“And when a patient gets a positive result, what they want to know is: *what’s the chance I’m actually sick?* That’s the **positive predictive value**.”

“That’s my precision. Forty percent.”

“And if the result is negative, the chance they’re really healthy is the **negative predictive value**. True negatives over everyone who tested negative.”

Léa scribbled. “Nine thousand seven hundred over nine thousand seven hundred and fifty. Ninety-nine and a half percent.”

$$\text{sensitivity} = \text{recall} = \text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{specificity} = \text{TN} / (\text{TN} + \text{FP}) = 1 - \text{FPR}$$

$$\begin{aligned}\text{FPR} &= \text{FP} / (\text{FP} + \text{TN}) \\ \text{PPV} = \text{precision} &= \text{TP} / (\text{TP} + \text{FP}) \\ \text{NPV} &= \text{TN} / (\text{TN} + \text{FN})\end{aligned}$$

“It’s one table,” Yannick said. “Everyone just reads it in a different direction. Doctors read the rows. Search engines read the columns. It’s still four boxes.”

. . .

The next morning Léa brought the napkin to work. Kerguelen smoothed it flat, read it, and nodded.

“And now the lazy model?”

“Recall zero. It catches nothing. Specificity a hundred percent: it never bothers anyone. And precision...” She hesitated. “It never raises an alarm, so it’s zero divided by zero. It doesn’t even exist.”

“A precision that doesn’t exist. That’s a model with nothing to say.” Kerguelen folded the napkin and put it in her drawer, next to the grey USB stick. “Now tell me something. Your model doesn’t really say ‘fraud’ or ‘honest’, does it? What does it actually give you?”

Léa thought. “A score. Between zero and one. I said ‘fraud’ when the score was over a half.”

“Why a half?”

Léa opened her mouth and closed it again.

“Every threshold you choose,” said Kerguelen, “gives you a different confusion matrix. Different precision, different recall, different everything. Which means you haven’t built one classifier. You’ve built a hundred of them, and you’ve looked at exactly one.” She reached for her coat. “Come on. I want to tell you about some radar operators.”

What Léa learned

The confusion matrix (TP, FN, FP, TN) is the basis of every metric. A cost matrix prices each kind of error, when you can get one. Precision = $TP/(TP+FP)$ is quality; recall = $TP/(TP+FN)$ is quantity; they trade off. F1 is their harmonic mean and ignores TN; F-beta shifts the weight (F2 favours recall, F0.5 precision). In medicine recall is sensitivity; specificity = $TN/(TN+FP) = 1 - FPR$; PPV = precision; NPV = $TN/(TN+FN)$.

The Radar Operators

The castle of Brest squats at the mouth of the Penfeld river, and inside it is the naval museum. Kerguelen walked Léa past the ship models and the figureheads without slowing down, until they reached a glass case holding a grey metal box with a round green screen.

“A radar console,” she said. “From the war. Imagine you’re nineteen, it’s three in the morning, and you’re watching this screen. A blip appears. It might be a bomber. It might be a flock of geese. It might be nothing but noise in the electronics. You have to decide, right now, whether to wake the whole coast.”

“And if you’re wrong?”

“Two ways to be wrong. Stay quiet when it’s a bomber, and people die. Raise the alarm for geese, and you’ve dragged a thousand men out of bed for nothing. Some operators were jumpy. They caught every plane and raised a hundred false alarms. Some were cautious. Hardly any false alarms, but they missed planes. The engineers realized they couldn’t judge a *receiver* — an operator with his machine — by one number. They had to look at the whole trade-off. They called it the **Receiver Operating Characteristic**. ROC.”

. . .

In a café by the castle walls, Kerguelen drew a square on a paper napkin.

“Horizontal axis: the false positive rate. What fraction of the honest customers do you wrongly flag? That’s the *cost*. Vertical axis: the true positive rate, your recall. What fraction of the frauds do you catch? That’s

the *benefit*. Every classifier, with its threshold fixed, becomes a single point in this square.”

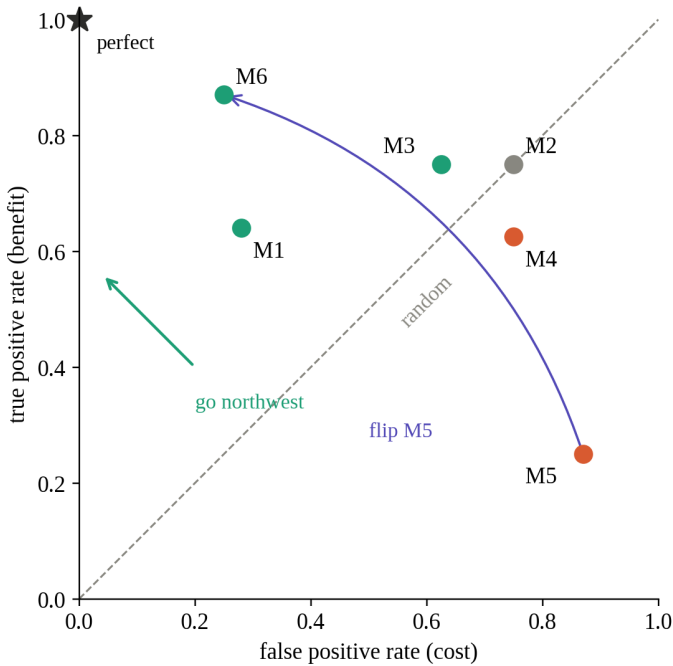
“Four corners,” she went on, marking them. “Bottom left, zero and zero: the model that never says fraud. No false alarms, no catches. Your lazy model lives here. Top right, one and one: the model that says fraud to everything. Catches every fraud, blocks every customer. Top left, zero and one: perfection. Every fraud caught, not one false alarm. Nobody lives there, but everybody’s trying to move toward it.”

She drew the diagonal from bottom left to top right.

“And this is chance. Suppose you ignore the transaction completely and flip a coin that says ‘fraud’ thirty percent of the time. You’ll flag thirty percent of the frauds and thirty percent of the honest customers. You land at thirty, thirty. Any coin you like lands somewhere on this line. So the diagonal is the line of *knowing nothing*.”

“So I want to be above it.”

“You want to go **northwest**. Up means catching more frauds; left means fewer false alarms. Northwest is better, always.”



The ROC square. Top left is perfect; the diagonal is random guessing. Six models from the lecture are plotted: M5 sits far below the diagonal, and flipping its answers gives M6.

...

Kerguelen pulled out a sheet of paper covered in confusion matrices. “Six models. Let’s put them on the map.”

“M1: catches sixty-four percent of frauds, false alarm rate twenty-eight percent. Accuracy sixty-eight percent. A decent point, well above the diagonal.”

“M2: true positive rate seventy-five, false positive rate seventy-five.”

Léa put it on the napkin. “It’s *on* the diagonal. It’s no better than a coin.”

“And its accuracy is exactly fifty percent on a balanced test. Now M5. True positive rate twenty-five percent, false positive rate eighty-seven. Accuracy nineteen percent.”

The dot landed deep in the bottom right, far below the diagonal. “That’s awful. It’s *worse* than random.”

“Is it? Think. To be that reliably wrong, what must it know?”

Léa stared at the dot. Then at M6, the last matrix: true positive rate eighty-seven percent, false positive rate twenty-five. Accuracy eighty-one.

“M6 is M5... flipped,” she said slowly. “Take every answer M5 gives and say the opposite, and you get M6.”

“A model far below the diagonal isn’t stupid,” Kerguelen said. “It has learned something real and it’s wearing its coat inside out. Turn the coat round. The truly useless models are the ones *on* the diagonal.”

. . .

“Last two. M3: true positive rate seventy-five percent, false positive rate sixty-two. Accuracy forty-five. M4: true positive rate sixty-two, false positive rate seventy-five. Accuracy fifty-five.”

Léa plotted them. “M3 is slightly above the diagonal. M4 is slightly below. So M3 is the better one. But M4 has the better accuracy?”

“Look at their test sets.”

Léa checked. M3 had been tested on four hundred frauds and sixteen hundred honest transactions. M4 on sixteen hundred frauds and four hundred honest ones.

“Accuracy changes with the mix,” she said. “Put a model on a test set with a different balance of classes, and its accuracy moves, even if the model itself is exactly as good.”

“Whereas the true positive rate is computed only among the frauds, and the false positive rate only among the honest customers. Each is a rate *inside its own class*. Change the proportion of frauds, and neither rate moves. That’s why people say ROC analysis is insensitive to class imbalance.” She tapped the napkin. “The point doesn’t care how many frauds are in your test set. Accuracy does.”

. . .

“Now,” Kerguelen said, “remember your score. Between zero and one. You said ‘fraud’ above a half. Each threshold is one point on this napkin. Slide the threshold, and the point moves.”

She took out a list she had clearly prepared: twenty transactions, ten frauds and ten honest, each with the model’s score, sorted from highest to lowest.

score	.90	.80	.70	.60	.55	.54	.53	.52	.51	.505
class	F	F	H	F	F	F	H	H	F	H
score	.40	.39	.38	.37	.36	.35	.34	.33	.30	.10
class	F	H	F	H	H	H	F	H	F	H

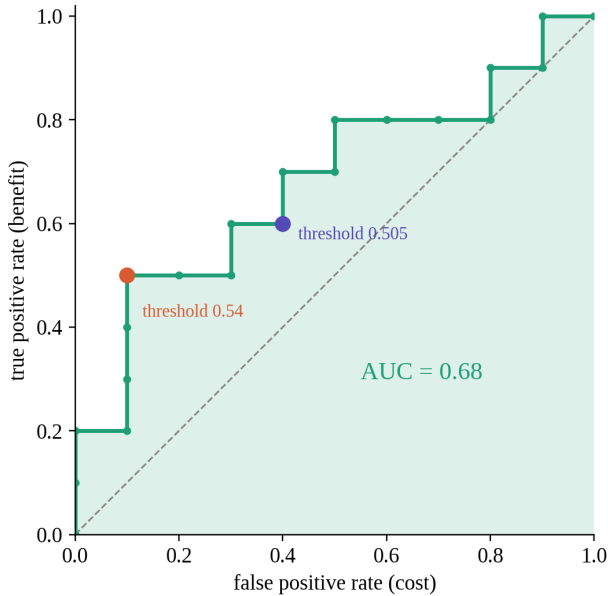
Twenty transactions sorted by the model's score (*F* = fraud, *H* = honest). This is the classic example from Fawcett's introduction to ROC analysis.

"Start with the threshold above point nine. Nothing is flagged: we're at zero, zero. Lower it to point nine. The top transaction is flagged. It's a fraud, so we move *up* one step: a tenth of the frauds caught. Lower it to point eight: another fraud, up again. Point seven: an honest customer. Now we move *right*: one tenth of the honest ones wrongly flagged. And so on, down the list. Every fraud, a step up. Every honest customer, a step to the right."

Léa traced it with her pencil. At point five four, she was at five frauds caught out of ten and one honest customer flagged: false positive rate point one, true positive rate point five. At point five oh five, near her old threshold of a half, she was at point four and point six. At the bottom of the list, point one, everything was flagged, and she arrived in the top right corner.

"It's a staircase," she said.

"It's a **ROC curve**. Not one classifier. All of them, every threshold at once. And to get it, the model has to give you scores, not just labels. You can draw a precision-recall curve the same way, if that's the trade-off you care about. When frauds are very rare, it's often the more honest picture, because precision *does* feel the imbalance: it's the one that tells the fraud team how much of their day will be wasted."



The ROC curve for the twenty scored transactions. Each fraud moves the curve up a step, each honest transaction moves it right. The shaded area under the curve, the AUC, is 0.68.

...

“How do I compare two whole curves?”

“The simplest way is the area underneath. The **AUC**, the area under the curve. A perfect model hugs the top left corner and has area one. A coin runs along the diagonal and has area one half. Yours is point six eight.”

“What does point six eight *mean*, though?”

Kerguelen smiled; she loved this one. “Pick one fraud at random, and one honest transaction at random. Ask the model which one is more suspicious. The AUC is the probability that it ranks the fraud higher. Ten frauds times ten honest customers makes a hundred pairs. Check them all. The model gets sixty-eight right.”

Léa did check them, on the back of the napkin, while Kerguelen ordered another coffee. Sixty-eight.

“The bigger the AUC, the better the model,” Kerguelen said. “On average. But be careful.”

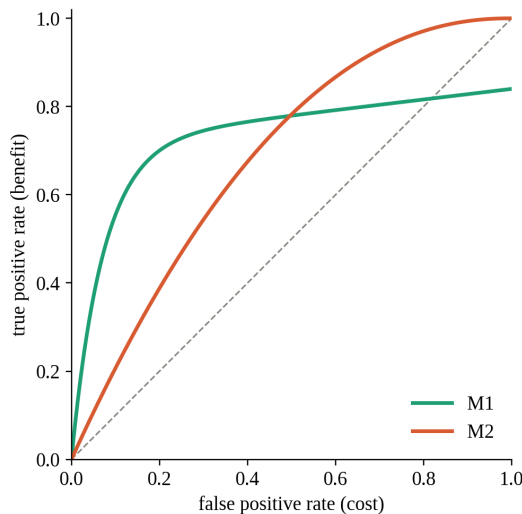
She drew two curves that crossed. One rose steeply from the corner and then flattened; the other rose slowly and finished higher.

“Neither model beats the other everywhere. The first is better when you can only afford a few false alarms, on the left. The second is better when you can tolerate many, on the right. So which one should the bank use?”

Léa thought of the fraud team: six people, a finite number of phone calls per day.

“The first. We can only afford a low false positive rate. So the only part of the curve that matters to us is the left edge.”

“Exactly. The AUC summarizes the whole curve. Your business lives on one part of it. Always ask where you’ll actually operate.”



Two crossing ROC curves. M1 is better at low false positive rates; M2 is better at high ones. Neither dominates, so the choice depends on where you will operate.

They walked back along the ramparts. The rain had stopped, for once, and the whole harbour glittered.

“I think I understand how to measure a model now,” Léa said.

“Good. Because on Monday, Jean-Yves wants a real system. And measuring one model is a very different thing from choosing between twenty.”

What Léa learned

ROC space plots TPR (benefit) against FPR (cost). (0,0) never says positive, (1,1) always does, (0,1) is perfect, and the diagonal is random guessing; aim northwest. A model below the diagonal can be flipped. Accuracy depends on the class mix while TPR and FPR do not. Sliding the threshold on scores traces a full ROC curve. AUC is the probability that a random positive is scored above a random negative (1 perfect, 0.5 random). When curves cross, choose the one that is better in your operating region.

Chapter Seven

So...

On Monday, Jean-Yves Le Bihan gave them two weeks.

“A fraud system that goes live in November. I don’t need it perfect. I need to know I can trust it.”

When he had gone, Kerguelen turned the whiteboard back to its first side. The big circle was still there, faded, with the dot of the truth outside it and h-star just inside the edge.

“Two weeks ago, you asked where errors come from. Now I’ll ask you the last question. How do you actually *get* to h-star?”

She drew five more dots inside the circle, scattered, labelled h1 to h5.

“You don’t find it by magic. You search for it. And a search needs five things.” She counted on her fingers, as she had on the first morning.

“One: a **family of models**. Your circle, H. Trees, logistic regression, whatever you choose. Two: a **parameter space**. The dials on that family — depth, the number of trees, a regularization strength. Three: a **search method** for trying candidates. Every combination on a grid, or random combinations, or something cleverer. Four: a **validation framework**. The protocol that estimates, honestly, how well each candidate will do on unseen data. And five: a **score function**. The single measure you’ll use to rank them. Change any one of the five and you may end up at a different dot.”

. . .

Léa spent three days on it. Her circle, to start with, held two families: logistic regression, because it was simple and the director would

understand it, and a forest of decision trees, because the literature said forests were good at fraud. Mathis added a gradient-boosted model, because Mathis always added a gradient-boosted model.

For the validation framework she used everything she'd learned. Stratified ten-fold cross-validation with a fixed seed, so every fold held its fair share of the rare frauds. Tuning inside the training folds only, never on the fold being tested. And, because fraud drifts, a second check: train on January to August, test on September and October. The grey USB stick stayed in Kerguelen's drawer.

For the score, she went back to the napkin. The fraud team could handle about two hundred alerts a day, so she cared most about the left side of the ROC curve. She settled on recall at a false positive rate of one percent, with the F2-measure alongside, and the cost matrix as the final judge whenever the numbers were close.

On Thursday the results came in. Logistic regression: recall of seventy-one percent at one percent false alarms. The forest: seventy-four. Mathis's boosted model: seventy-four point four.

"I win," said Mathis.

. . .

"Do you?" said Kerguelen.

"Point four higher."

"On what?"

"The average over the ten folds."

"And the spread?"

Mathis checked. His model's fold scores ranged from sixty-nine to seventy-nine. So did the forest's.

"When two averages differ by less than the wobble inside each of them," Kerguelen said, "you haven't shown a difference. You've shown noise. To claim that one model beats another, you need a **statistical test**. Compare them fold by fold, on exactly the same folds, and ask how likely a difference this size would be if the two models were really equally good. There are standard tools for that — a paired t-test, or the Wilcoxon signed-rank test when you don't trust the numbers to be well-behaved. They have their own subtleties, because cross-validation folds share

training data, but at least they force you to ask the question.”

Mathis ran the test. The difference between his model and the forest was nowhere near significant.

“And compared with logistic regression?”

The forest’s advantage over the simple model held up, fold after fold, and in the September–October check too.

“Then here’s how I’d put it,” Kerguelen said. “The forest and the boosted model are indistinguishable on our data. Both genuinely beat logistic regression. Of the two, the forest is simpler to maintain, faster to retrain, and easier to explain.”

“Occam,” said Léa.

“Occam.”

. . .

“There’s one more thing,” Kerguelen said. “You’ve compared three models on one bank’s data. If you ever want to claim that one *algorithm* is better than another in general — the kind of thing people write papers about — the bar is higher. Your **datasets** must be representative and varied: different banks, different countries, different years. Your **protocol** must be appropriate and identical for every candidate. Your **measures** must match the goal, and there’s rarely just one: accuracy, precision, recall, AUC, cost, speed, how complex the model is. And your conclusions must survive the statistical tests.” She shrugged. “Remember Wolpert and Macready. Somebody’s free lunch is always somebody else’s bill.”

What Léa learned

Reaching h^* is a search that needs five ingredients: a model family (H), a parameter space, a search method, a validation framework and a score function. To compare models: use representative and varied datasets, the same appropriate protocol for all, measures chosen for the goal, and statistical tests before claiming a winner. When candidates are statistically tied, prefer the simpler one.

Epilogue

Ninety-Eight Percent

The presentation was on the top floor, in the room with the good chairs.

Léa showed Jean-Yves the confusion matrix first, with the four boxes named. Then the cost: roughly two hundred thousand euros a month saved compared with doing nothing. Then the ROC curve, with a small red dot where the system would operate, at one percent false alarms, where the fraud team could keep up. Then the cross-validation results, the average and the spread, and the out-of-time test on September and October. And then, last, the grey USB stick, opened for the first time that morning, which had given almost exactly the same numbers.

Jean-Yves listened to all of it, nodding. At the end he took his glasses off.

“Very good. Just one question. What’s the accuracy?”

Léa glanced at Kerguelen, who was examining the ceiling with great interest.

“Ninety-eight percent,” Léa said. “Which is lower than a model that does absolutely nothing.”

There was a pause. Then Jean-Yves laughed, and the system went live in November.

. . .

In January a new intern arrived: Gaël, twenty-two, with a laptop he carried open in front of him like a waiter carrying a cake.

“Ninety-nine percent!” he announced.

Léa didn't look up. She had taken the wobbly chair for her own desk, and she found she rather liked it.

"Ninety-nine percent," she said, "of what?"

Cheat Sheet

Concept	Meaning	Ch.
Generalization error	$\Pr_{(x,y) \sim P}[h(x) \neq y]$: the chance of being wrong on a new example.	2
Error decomposition	total error = bias + variance + ϵ (irreducible noise).	2
Bias	Gap between the truth f and the best average model h^* in H . Too-simple models: underfitting.	2
Variance	Gap between your model h and h^* . Depends on the sample. Too-complex models: overfitting.	2
Loss $\ell(y, y')$	0 if $y = y'$, positive otherwise. Absolute $ y - y' $, squared $(y - y')^2$, zero-one $1(y \neq y')$, linear $\frac{1}{2}(1 - y \cdot h(x))$.	3
True risk	$R_P(h) = E[\ell(h(x), y)]$ over P . Never computable.	3
Empirical risk	$R_S(h) = (1/n) \sum \ell(h(x_i), y_i)$. Optimistic for the model you trained.	3
Holdout	Train on about 2/3, test on 1/3; repeat with different shuffles.	4
Train / tune / test	Choose settings on the tune set; touch the test set once.	4
k-fold CV	k parts (usually 10); each is the test set once; report mean and spread.	4
Leave-one-out	$k = n$; for very small datasets.	4

Concept	Meaning	Ch.
Splitting	Size, shuffle, stratify, fix the seed (and respect time order when needed).	4
Overfitting	h' : $\text{error}_{\text{train}}(h) < \text{error}_{\text{train}}(h')$ but $\text{error}_p(h) > \text{error}_p(h')$.	4
Accuracy / error	$(TP + TN)/n$ and $(FP + FN)/n$. Misleading with imbalanced classes.	1, 5
Precision (PPV)	$TP / (TP + FP)$. When it says positive, how often is it right?	5
Recall, sensitivity, TPR	$TP / (TP + FN)$. Of the real positives, how many were found?	5
Specificity	$TN / (TN + FP) = 1 - \text{FPR}$.	5
False positive rate	$FP / (FP + TN)$. The false alarm rate.	5
NPV	$TN / (TN + FN)$.	5
F1	$2pr / (p + r) = 2TP / (2TP + FN + FP)$. Ignores TN.	5
F-beta	$(1 + \beta^2) pr / (\beta^2 p + r)$. $\beta > 1$ favours recall, $\beta < 1$ precision.	5
ROC space	TPR (y) against FPR (x). (0,1) perfect, diagonal random, go northwest.	6
AUC	Area under the ROC curve = $P(\text{random positive scored above random negative})$. 1 perfect, 0.5 random.	6
Model search	Model family, parameter space, search method, validation framework, score function.	7
Comparing models	Representative data, same protocol, goal-driven metrics, statistical tests.	7

Appendix

Map to the Lecture

Chapter	Lecture	Covers
1. Ninety-Nine Percent	Section 1, Issues	accuracy trap, train vs test, what makes a good pattern
2. The Three Sources of Error	Section 2, Bias & variance	setup, inductive bias, PAC, bias–variance, polynomial example, Occam’s razor
3. The Price of a Mistake	Section 2 (end)	loss functions, true and empirical risk, disagreement
4. The Vault	Section 3, Protocols	holdout, train/tune/test, cross-validation, LOO, splitting, overfitting, No Free Lunch
5. Four Boxes	Section 4, Metrics	confusion and cost matrices, precision, recall, F-measures, sensitivity, specificity
6. The Radar Operators	Section 5, ROC	ROC space, models M1–M6, ROC curve from scores, AUC, crossing curves
7. So...	Section 6, So...	exploring H, comparing models, statistical tests

Further reading

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